Day 2: TCP/IP SuiteTheory

Tasks : 1 Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

To capture network traffic while refreshing the Flipkart homepage and identify the four TCP/IP layers by examining packet headers in Wireshark.

Procedure

1. Open Wireshark and select the active network interface.
2. Start capturing packets and refresh the Flipkart homepage in the browser.
3. Stop the capture after a few seconds.
4. Examine the Protocol and Info columns in the packet list.
5. Select packets and inspect their headers to identify the TCP/IP layers.

Packet Analysis

TCP/IP Layer	Frame No.	Protocol / Header	Observation
Application	1471	TLS	The packet contains data identified as a TLS record, indicating encrypted application traffic.
Transport	4	TCP	The TCP header contains source port 56791 and destination port 443.
Internet	4	IPv4	The IPv4 header shows source IP 192.168.1.6 and destination IP 162.247.241.14.
Network Access	29	Ethernet II / ARP	The frame contains an ARP request used to find the MAC address associated with an IPv4 address on the local network.

Observations

- **Application Layer:** Frame 1471 contains TLS-formatted data. HTTPS traffic uses encryption to protect communication.
- **Transport Layer:** Frame 4 uses TCP, with destination port 443, commonly associated with HTTPS traffic.
- **Internet Layer:** Frame 4 contains IPv4 addresses that identify the source and destination of the packet.
- **Network Access Layer:** Frame 29 contains an ARP request and Ethernet addressing information used for communication on the local network.

Conclusion

Wireshark helped identify the four TCP/IP layers by examining application data, transport protocols, IP headers, and local network frames. The capture contains TCP, UDP, IPv4, IPv6, ARP, and encrypted TLS traffic.

Task 2 : Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

To compare the OSI and TCP/IP models and understand their layers with real-world examples.

Comparison Table

OSI Model	TCP/IP Model	Function	Real-World Example
Application (Layer 7)	Application	Provides network services to applications.	Opening Flipkart in Chrome or sending a WhatsApp message.
Presentation (Layer 6)	Application	Handles data format, encryption, and compression.	Encrypted HTTPS communication or displaying a JPEG image.
Session (Layer 5)	Application	Helps establish, manage, and end communication sessions.	Maintaining a login session during online shopping.
Transport (Layer 4)	Transport	Uses TCP/UDP and port numbers to support communication between applications.	TCP traffic to a website or UDP traffic during a voice call.
Network (Layer 3)	Internet	Uses IP addresses and routing to move packets between networks.	A router forwarding a packet toward a server.
Data Link (Layer 2)	Network Access	Handles frames and local network addressing.	Wi-Fi or Ethernet communication between nearby network devices.
Physical (Layer 1)	Network Access	Transmits bits through physical signals or wireless radio waves.	Wi-Fi radio signals, Ethernet cables, or fiber-optic cables.

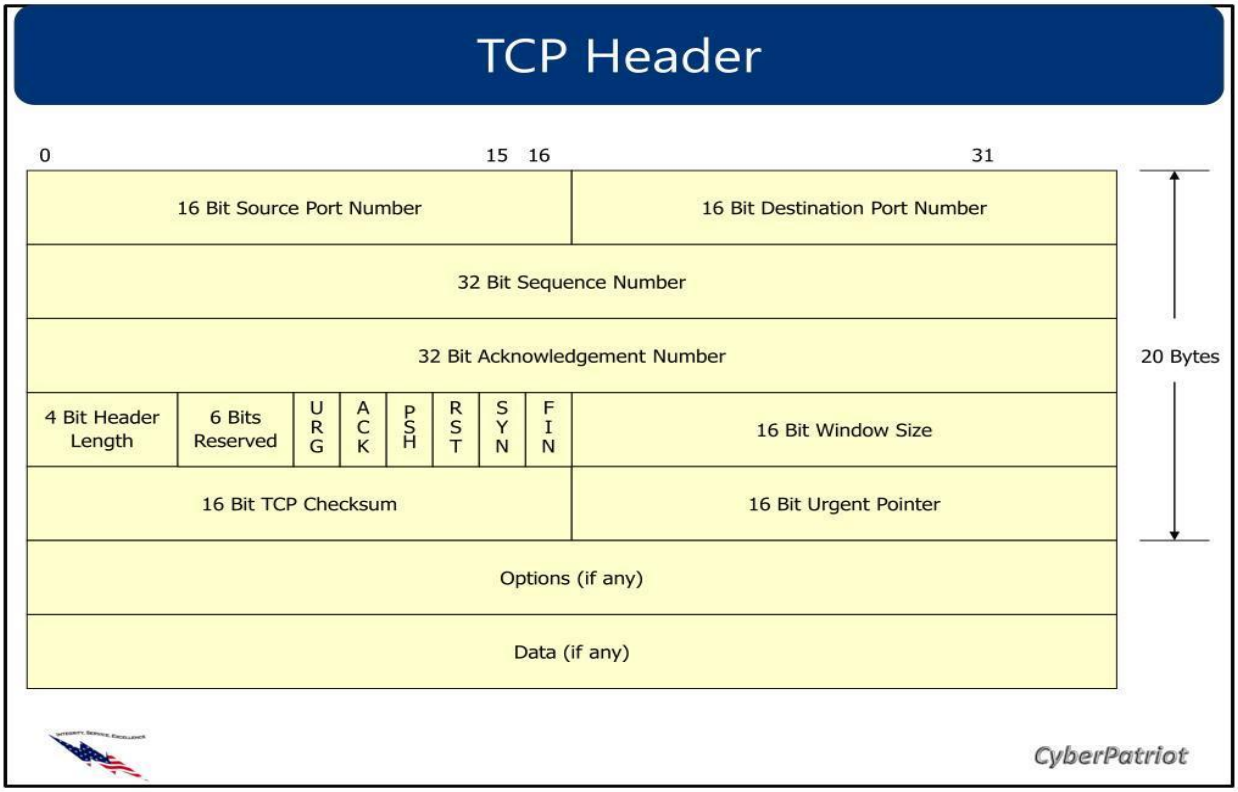
Main Differences

1. The OSI model contains seven layers, while the TCP/IP model contains four layers.
2. The OSI model separates Application, Presentation, and Session layers. TCP/IP combines their functions into its Application layer.
3. The OSI model separates Data Link and Physical layers. TCP/IP combines their functions into its Network Access layer.
4. TCP/IP is the practical protocol suite used for communication across the internet, while OSI is commonly used as a reference model for learning and troubleshooting networking.

Conclusion

Both models explain how data travels between devices over a network. The OSI model provides a more detailed seven-layer structure, while the TCP/IP model groups related functions into four layers. Real-world applications such as WhatsApp and Flipkart rely on multiple layers working together to communicate successfully.

Task 3 : Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.



To identify the important fields of a TCP segment header and understand how they help in reliable communication between devices.

Introduction

TCP stands for **Transmission Control Protocol**. It is a Transport Layer protocol that provides reliable and ordered data delivery between applications.

A TCP segment contains a header and the data being transmitted. The header carries control information, such as port numbers, sequence numbers, acknowledgment numbers, and flags.

TCP Header Fields and Their Functions

1. Source Port

The Source Port identifies the port number of the sending application. It helps the receiving device identify the application endpoint associated with the communication.

Example: A computer may use a temporary port such as 56791 when connecting to a website.

2. Destination Port

The Destination Port identifies the service or application endpoint on the receiving device.

Example: Port 443 is commonly used for HTTPS web traffic.

3. Sequence Number

The Sequence Number helps TCP keep track of the order of bytes sent during communication. It allows the receiver to identify missing or out-of-order data.

Example: If data arrives out of order, TCP can use sequence numbers to arrange the bytes correctly.

4. Acknowledgment Number

The Acknowledgment Number indicates the next byte the receiver expects to receive when the ACK flag is set. It helps confirm which data has been received.

Example: An acknowledgment number of 501 generally means the receiver expects byte 501 next.

5. Header Length (Data Offset)

This field specifies the length of the TCP header and indicates where the actual data begins. A TCP header without options is normally 20 bytes long.

6. Flags (Control Bits)

Flags control important TCP operations, such as establishing, managing, and closing a connection.

Common flags include:

- **SYN:** Used to establish a TCP connection.
- **ACK:** Indicates that the acknowledgment field is valid.

- **FIN:** Used to close a TCP connection normally.
- **RST:** Resets a connection.
- **PSH:** Requests that received data be passed to the application promptly.
- **URG:** Indicates that the Urgent Pointer field is significant.

7. Window Size

The Window Size indicates how much additional data the receiver is currently willing to accept. TCP uses this information for flow control, helping prevent a sender from overwhelming the receiver.

8. Checksum

The Checksum helps detect corruption in the TCP header and data during transmission. If corruption is detected, TCP can use its reliability mechanisms to handle the problem.

9. Urgent Pointer

The Urgent Pointer indicates the position associated with urgent data when the URG flag is set. It is used in conjunction with the Urgent flag.

10. Options

The Options field carries additional TCP settings and capabilities. Examples include the Maximum Segment Size (MSS), window scaling, and timestamps.

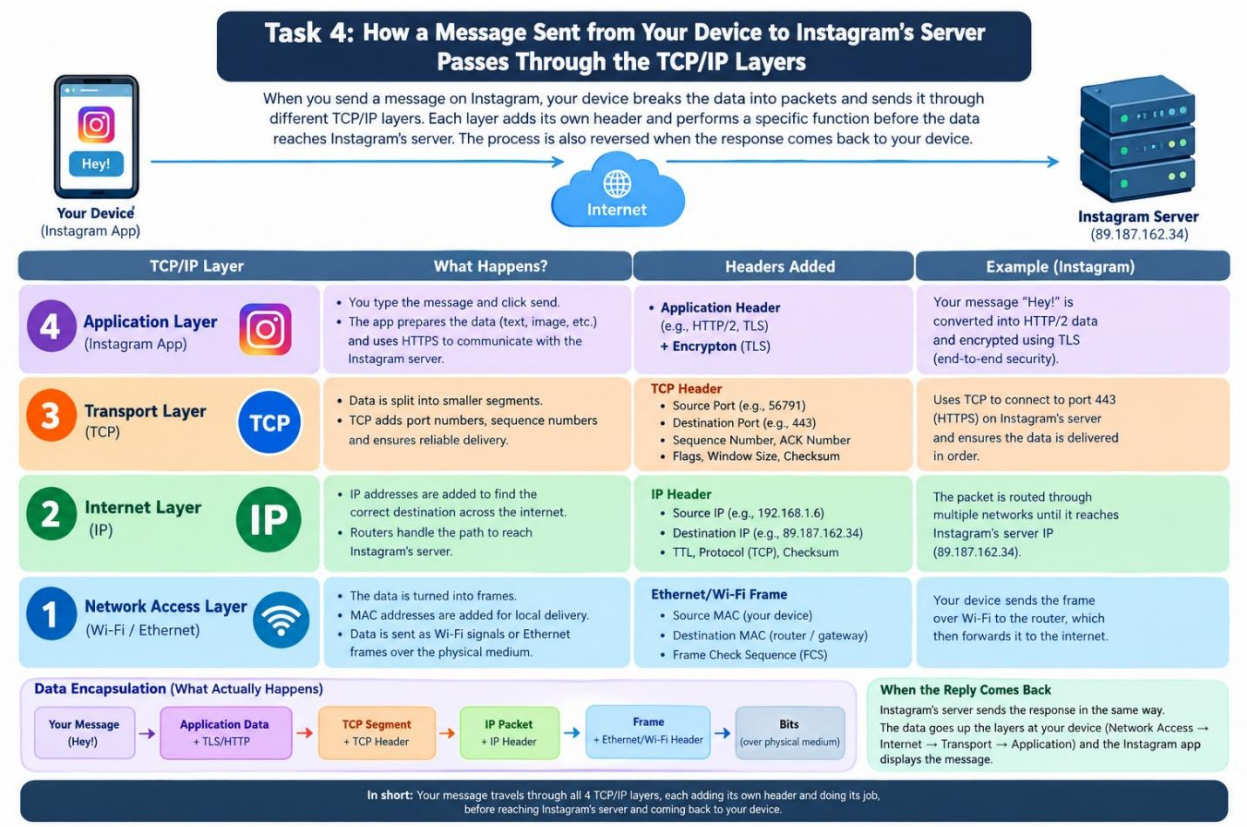
11. Data (Payload)

The Data field contains the application data being transmitted. It is not part of the TCP header itself, but it follows the header within the TCP segment.

Conclusion

The TCP segment header contains information required to manage communication between applications. Port numbers identify communication endpoints, sequence and acknowledgment numbers help track data, flags control connections, and the window size supports flow control. Together, these fields help TCP provide reliable and ordered data delivery.

Task 4 : Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).



Conclusion: When I send a message through Instagram, it passes through all four TCP/IP layers. Each layer performs a specific function, such as preparing application data, adding TCP information, adding IP addresses, and transmitting data over Wi-Fi or Ethernet. At the destination, the data is processed in reverse order and delivered to the application.

Task 5 : Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

To analyze a captured packet, identify its TCP/IP layer, and explain its possible role in real-world network communication.

Selected Packet Details

- **Frame Number:** 4
- **Protocols:** TCP and IPv4
- **Source IP Address:** 192.168.1.6
- **Destination IP Address:** 162.247.241.14
- **Source Port:** 56791
- **Destination Port:** 443

TCP/IP Layer Identification

The selected packet contains both TCP and IPv4 headers.

Transport Layer: The TCP header contains source port 56791 and destination port 443. TCP supports communication between application endpoints and provides mechanisms for reliable data delivery.

Internet Layer: The IPv4 header contains source and destination IP addresses. These addresses help identify the packet's source and destination across networks.

How Was the Packet Identified?

I examined the packet in Wireshark and checked its Protocol column and packet details. The TCP header showed the port numbers, while the IPv4 header showed the source and destination IP addresses. Therefore, the packet helped me identify both the Transport and Internet layers.

Real-World Application

The capture was taken while browsing and refreshing the Flipkart homepage. TCP traffic using destination port 443 is consistent with HTTPS web communication. This type of traffic may support secure website loading and data exchange between a browser and a web server.

However, the packet details alone do not confirm that this specific packet belongs to Flipkart or represents a particular webpage request.

Conclusion

This packet demonstrates how TCP and IPv4 work together during network communication. TCP provides transport-related information, while IPv4 provides addressing information for routing packets across networks. Wireshark makes it possible to examine these headers and understand how internet communication works.